

When Do Parts Become a Whole?

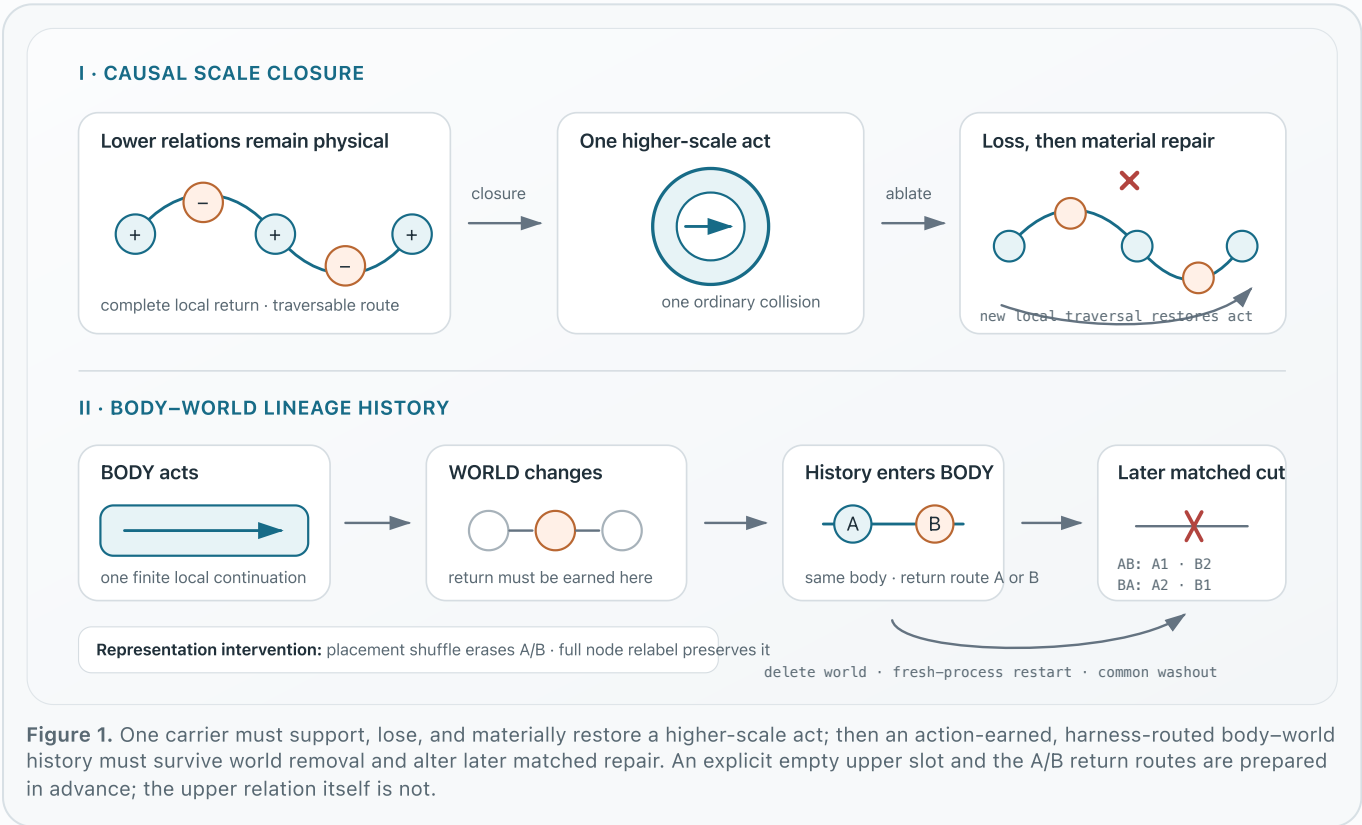
A minimal non-neural test of causal scale closure and material lineage memory

Amir Tlinov Independent research July 2026 Code & evidence ↗

THE QUESTION AND HYPOTHESIS

Regeneration and morphogenesis make an old question experimentally concrete: **when do locally interacting parts become one higher-scale causal individual, rather than merely an aggregate?** Work on bioelectric pattern memory and multiscale competency shows that tissues can retain history, correct anatomical error, and pursue large-scale outcomes despite perturbation [1–3]. This work does not model planarian physiology. It asks for a narrower, substrate-independent minimum that can be built, injured, and falsified.

The specific contribution is a joint intervention test: the same local-rule carrier must acquire a higher-scale act, lose and rebuild it after ablation, and carry an action-earned, harness-routed body–world history into later matched repair. Contact becomes meaningful *for a body* here only when it leaves a persistent relation in that body and changes what it can do after a later perturbation.



OPERATIONAL CRITERION

A bounded causal scale closure must survive interventions, not merely resemble a whole.

Causal octave (1:2) is the informal name for retained differences acting as one new causal unit without erasing lower causes—not a numerical ratio.

1. Lower relations remain materially present and traversable.
2. The group supports one act unavailable before closure.
3. Targeted ablation removes the act while preserving the lower route.
4. A new local traversal through surviving material restores it.
5. A world-earned relation survives world removal and changes response to a later matched injury.

1 rule

collision kernel SHA-256
0d32047eecb3...

THE CARRIER

The model stores persistent local links between fixed records; one immutable collision rule updates one linked encounter at a time. The harness initializes the body, removes a declared link, supplies the world interaction, and measures recovery. An explicit empty upper slot and the A/B return routes are pre-installed; the upper relation itself is not. No reward, target evaluator, planner, learned model, or score-dependent update participates in the carrier's state transition. Frozen audit: 21 tests · world-lineage executable source SHA-256: 2ac3304f0a95...

Scale closure · ablation · restoration

Closure and restart passed after growth spans 1, 7, 127, and 1023 (closed cycles: 6, 12, 132, 1028 places). The full court used one 12-place lower cycle: after the lower route closed, its earned upper relation gave a one-collision PASS; removing only that relation gave HALT while the cycle remained traversable; one new traversal rebuilt the relation at collision 12 and restored PASS. A separate 18-record address-relabel control preserved trace and outcome.

World-mediated relational history

A 1024-record body acted through a five-record world using the same record format and rule; return was valid only from a world record changed by that passage. The harness directed the earned final return into prebuilt route A or B. After world deletion, fresh-process restart, identical washout, and the same cut, A restored the upper entry in episode 1 of AB and 2 of BA; B gave the mirror pattern. Shuffling all 257 return-relation placements erased A/B; full node relabel preserved it.

A1 · B2

upper-entry birth episode in
AB; mirrored in BA

4 / 4

prespecified deterministic
initializations; not
independent replicates

112

paired G1/G2 world-
continuity controls

8 + 8

separate placement-shuffle
+ full-relabel interventions

What the result supports

A constructive existence result in an engineered carrier: persistent local relations can support a bounded higher-scale act, lose it under targeted injury, rebuild it through surviving material, and retain body-world history that changes later repair. Shuffle versus relabel identifies a distribution-dependent relation class; it does not localize memory to a small site.

What it does not support

It is not evidence that natural matter arises from differences, not a model of bioelectric physiology, and not basal cognition. Topology, upper slot, and A/B routes are specified. The five-record world is the same engineered physics, not a distinct material class, and the harness selects its final return route. Pre-arranged repair architecture remains the strongest alternative; a biological analogue must also rule out incomplete washout.

BIOLOGICAL QUESTION

Would this reveal where lineage history is carried in tissue: impose two bioelectric histories, wash out until a prespecified measurement panel matches, apply the same injury, test for divergent repair, then perturb the candidate state that predicts it? **Which measurement and intervention would best distinguish causally effective lineage memory from incomplete washout or a pre-wired circuit?** This need not oppose a distributed morphogenetic setpoint; it asks which physical relations carry history.

REFERENCES

1. Levin, M. (2023). *Bioelectric networks: the cognitive glue enabling evolutionary scaling from physiology to mind*. *Animal Cognition* 26, 1865–1891. doi:10.1007/s10071-023-01780-3
2. Levin, M., Pietak, A. M., & Bischof, J. (2019). *Planarian regeneration as a model of anatomical homeostasis*. *Seminars in Cell & Developmental Biology* 87, 125–144. doi:10.1016/j.semcdb.2018.04.003
3. Levin, M. (2022). *Technological Approach to Mind Everywhere*. *Frontiers in Systems Neuroscience* 16, 768201. doi:10.3389/fnsys.2022.768201